

Topographic Radar Geometry-Guided Cross-Attention for Alpine Avalanche Debris Mapping in Sentinel-1 SAR: Cross-System Generalization Across the Alps, Pamirs, and Arctic

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- Akshay Sharma: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.
- Lalji Prasad: Supervision, Methodology, Resources, Validation, Writing – Review & Editing.

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The AvalCD benchmark dataset is publicly accessible on Zenodo (DOI: [10.5281/zenodo.15863589](https://doi.org/10.5281/zenodo.15863589)). All source code, neural network implementations, evaluation scripts, and replication suites are publicly accessible at <https://github.com/akssha74/TopoRadar-Net-AvalCD>. Publication-bound TopoRadar-Net and Attention U-Net checkpoints for all three seeds and all publication artifacts are distributed under immutable Release v1.0.7: <https://github.com/akssha74/TopoRadar-Net-AvalCD/releases/tag/v1.0.7>.